

Exam Date & Time: 14-Jun-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester Makeup Examination, June -2025

CRYPTOGRAPHY AND NETWORK SECURITY [IT-4001]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

1) Consider the following Playfair array.

A)

B	A	R	M	G
U	E	I/J	T	N
H	O	S	D	W
Y	L	P	C	F
K	Q	V	X	Z

(5)

I. Encrypt the plaintext happy days.

II. Decrypt the ciphertext TERCSUBW

B) A) Identify the security attack against which the following mechanisms give protection.

- Traffic Padding
- Notarization
- Access Control

(5)

B) Define the type of security attack in each of the following cases:

- A student breaks into a professor's office to obtain a copy of the next day's test.
- A student sends hundreds of e-mails per day to another student using a phony return e-mail address.

2) Raj uses ElGamal for creating a secure system which projects Confidentiality, Integrity, Nonrepudiation and Authentication. Given the prime $p=13$, $e_1 = 2$, $r=5$ and $d=3$.

A)

- Encrypt the message "11".
- Encrypt the cipher text to get back the plain text.
- Calculate the two signatures S_1 and S_2 .
- how the verification process for the above calculated signatures.

(5)

Show with suitable justifications how Confidentiality, Integrity, Nonrepudiation and Authentication is maintained in the above communication scenario.

- B) Alice and Bob were two cryptographers tasked with establishing a secure communication channel across an unsafe network using the Diffie-Hellman key exchange protocol. They employ the prime number p : 97 and the Generator number α : 2. The private keys in question are Bob's private key b : 27 and Alice's private key a : 13. Calculate their common secret key. Also, justify why Diffie-Hellman key exchange is more secure than traditional methods of key exchange. (5)
- 3) Distinguish between Matyas-Meyer-Oseas Scheme and Miyaguchi-Preneel Scheme with neat diagrams. (5)
- A)
- B) What is ECC Cryptography? What are the benefits of ECC? Given an elliptic curve and two points $P = (1, 1)$ and $Q = (2, 7)$ on the curve, calculate the point $P + Q$ using the group law. The Elliptic Curve: $y^2 = x^3 - 3x + 5 \pmod{23}$. (5)
- 4) Illustrate how an attack can compromise a password-based authentication system in IP security. With a suitable example, show how this issue can be resolved using a Challenge-Response Authentication system. (5)
- A)
- B) Assess the role of firewalls in network security. What are the primary functions of a firewall, and how do they contribute to protecting an organization's network? (5)
- 5) Demonstrate the response of CSSV in a real time attack. Is it feasible for an intruder to employ an exhaustive computer search to discover the session key within given time? (5)
- A)
- B) Consider a MAANG institution that handles large amounts of sensitive employee information and faces daily cyber threats. In this situation, evaluate the important role of SSO? How effectively SSO can be used. (5)

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